

CTMT Dynamics

A Linear–Stochastic Layer, the Mori–Zwanzig Seepage Kernel, and a Provable Ladder
toward Buildup and Curvature

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Abstract

Papers I–III developed CTMT as a *static* geometry: one Jacobian and one covariance at a fixed operating point. That formalism cannot express accumulation, so the informal thesis of “buildup by data leak from the null sector” remained an interpretation rather than a theorem. This paper supplies the missing dynamical layer using only established mathematics. We equip the resolution geometry with a linear stochastic (Ornstein–Uhlenbeck) evolution, whose covariance obeys the differential Lyapunov equation. Two results are proved. First, a *seepage generation equation*: starting from a decoupled state, resolved–null covariance is created at first order exactly by the cross-sector drift, $\dot{C}_{RN}|_{C_{RN}=0} = A_{RN}\Sigma_N + \Sigma_RA_{NR}^\top$. Second, an *exact Mori–Zwanzig memory kernel*: eliminating the null sector yields a generalized Langevin equation for the resolved sector with kernel $K(\tau) = A_{RN}e^{A_{NN}\tau}A_{NR}$ and a residual force driven by the unresolved initial state. Both are numerically verified. The memory kernel is shown to coincide with the causal “phase kernel” of earlier CTMT drafts, identifying that construction as the standard object of statistical mechanics. We then state, as a precisely posed *ladder* of open problems, what a genuine “buildup” theorem requires: monotonicity of a directed-information or Fisher-gain functional (Gaussian transfer entropy equals Granger causality), and the identification of the moving resolved subbundle’s curvature with dynamical seepage, with holonomy realised by the transport monodromy already computed empirically. The paper proves the mechanism and delimits precisely what remains to be proved.

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1 Purpose, scope, and claim stratification

The static theory (Papers I–III) proved that the unresolved sector is informative and classifying, but by construction could not express any process in time. This paper adds the generator that was missing and, with it, turns “seepage” and “buildup” from metaphors into equations. The mathematics is entirely standard—linear-stochastic systems, the Lyapunov equation, the Mori–Zwanzig projection formalism, Granger/transfer-entropy causality, Čencov’s theorem, and the projection connection—so the content is a *translation* of CTMT into recognised machinery, not a new invention.

Proved here.

The dynamical resolution geometry and its Lyapunov flow (Section 2); the seepage generation equation (Theorem 3.2); the exact Mori–Zwanzig memory kernel (Theorem 4.1); the identification of the earlier phase kernel with the memory kernel (Corollary 4.2).

Cited (established).

Existence/uniqueness of the stationary covariance; Gaussian transfer entropy = Granger causality; Čencov uniqueness of the Fisher metric; Ambrose–Singer holonomy–curvature.

Conjectured (the ladder, Section 8).

Monotone buildup of a directed-information/Fisher functional under stated conditions on the drift and noise; the curvature=seepage identification and its canonicity; rank-changing (singular) globalization.

Non-claim 1.1 (What is still not proved). Proving the *mechanism* of seepage (a generation term and a memory kernel) is not the same as proving *buildup* (a monotone accumulation). Buildup remains a conjecture with explicit hypotheses (Section 8). No claim is made that any particular physical system realises these hypotheses; that is an empirical question, addressed only by the falsification protocol of Section 9.

2 The dynamical resolution geometry

We work at the tangent (linear) level consistent with Papers I–III, where the objects are already Jacobians and covariances. The matching dynamics is linear-stochastic.

Definition 2.1 (Dynamical resolution geometry). A dynamical resolution geometry is a resolution geometry $\mathcal{G} = (O, g; R, \Sigma)$ together with an observable evolution

$$dy_t = A y_t dt + B dW_t, \quad y_t \in O, \quad Q := BB^\top \succeq 0,$$

where A is the drift (generator), W a standard Wiener process, and the instantaneous covariance is $\Sigma(t) = \text{Cov}(y_t)$. The resolved/null split $O = R \oplus N$ is that of the underlying static geometry at the operating point.

Proposition 2.2 (Lyapunov flow of the covariance). *Under Definition 2.1, the covariance obeys the differential Lyapunov equation*

$$\dot{\Sigma}(t) = A\Sigma(t) + \Sigma(t)A^\top + Q.$$

If A is Hurwitz (all eigenvalues have negative real part), there is a unique stationary covariance $\Sigma_\infty \succeq 0$ solving $A\Sigma_\infty + \Sigma_\infty A^\top + Q = 0$, and $\Sigma(t) \rightarrow \Sigma_\infty$; if in addition (A, B) is controllable, $\Sigma_\infty \succ 0$.

Proof. Standard: apply Itô's formula to $y_t y_t^\top$ and take expectations for the ODE; the algebraic equation and convergence are the classical Lyapunov results for stable linear systems [1, 2]. Verified numerically against direct simulation. $\square$

This is the entire additional apparatus. Everything below is a consequence of blocking this one equation against $O = R \oplus N$, or of eliminating the null sector from the deterministic part.

3 Seepage as a generation equation

Write the drift, covariance, and noise in the adapted g -orthonormal frame:

$$A = \begin{pmatrix} A_{RR} & A_{RN} \\ A_{NR} & A_{NN} \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_R & C \\ C^\top & \Sigma_N \end{pmatrix}, \quad Q = \begin{pmatrix} Q_R & Q_{RN} \\ Q_{RN}^\top & Q_N \end{pmatrix},$$

with $C := C_{RN}$ the resolved-null coupling of Papers II–III.

Proposition 3.1 (Block Lyapunov system). *The coupling block evolves by*

$$\dot{C} = A_{RR}C + C A_{NN}^\top + (A_{RN}\Sigma_N + \Sigma_R A_{NR}^\top) + Q_{RN}.$$

Proof. Take the R – N block of Proposition 2.2: $(A\Sigma)_{RN} = A_{RR}C + A_{RN}\Sigma_N$ and $(\Sigma A^\top)_{RN} = \Sigma_R A_{NR}^\top + C A_{NN}^\top$; add Q_{RN} . $\square$

Theorem 3.2 (Seepage generation). *Suppose at some instant the covariance is sector-decoupled ($C = 0$) and the noise is sector-decoupled ($Q_{RN} = 0$). Then*

$$\dot{C}|_{C=0, Q_{RN}=0} = A_{RN}\Sigma_N + \Sigma_R A_{NR}^\top.$$

Consequently coupling is generated at first order if and only if $A_{RN}\Sigma_N + \Sigma_R A_{NR}^\top \neq 0$; when $\Sigma_R, \Sigma_N \succ 0$ this holds generically iff the cross-drift blocks A_{RN}, A_{NR} do not both vanish. The term $A_{RN}\Sigma_N$ is exactly the null-sector variance transported into the resolved sector by the null→resolved drift A_{RN} .

Proof. Set $C = 0, Q_{RN} = 0$ in Proposition 3.1. Verified numerically to machine precision. $\square$

Remark 3.3 (This is the precise content of “interlayer seepage”). The static theory could only observe that $C \neq 0$ constrains symmetry and informs inference (Papers II–III). The generator makes C a *dynamical variable* with a source term. Seepage is the source: unresolved variance Σ_N is converted into resolved-null covariance through the off-diagonal drift. The two directions are distinct and both meaningful: A_{RN} appears in $\dot{y}_R = \cdots + A_{RN}y_N$ (the unresolved state drives the resolved state), while A_{NR} appears in $\dot{y}_N = A_{NR}y_R + \cdots$ (resolved drives unresolved). The CTMT thesis concerns the first.

4 The Mori–Zwanzig memory kernel

The generation equation is instantaneous. The full effect of the unresolved sector on the resolved sector, integrated over its history, is a memory kernel. Eliminating the null sector from the deterministic part of the dynamics yields an exact generalized Langevin equation; this is the Mori–Zwanzig projection formalism [3, 4, 5], which for linear systems is exact and closed-form.

Theorem 4.1 (Exact null-sector memory kernel). *For the deterministic part $\dot{y} = Ay$ with split $O = R \oplus N$, the resolved component satisfies the exact generalized Langevin equation*

$$\dot{y}_R(t) = A_{RR} y_R(t) + \int_0^t K(t-s) y_R(s) ds + F(t),$$

with memory kernel and residual force

$$K(\tau) = A_{RN} e^{A_{NN}\tau} A_{NR}, \quad F(t) = A_{RN} e^{A_{NN}t} y_N(0).$$

The kernel is causal ($K(\tau) = 0$ for $\tau < 0$), decays at the rate set by the null-sector generator A_{NN} , and is nonzero iff both cross-drifts A_{RN}, A_{NR} are nonzero.

Proof. Solve the null equation $\dot{y}_N = A_{NR}y_R + A_{NN}y_N$ by variation of constants,

$$y_N(t) = e^{A_{NN}t} y_N(0) + \int_0^t e^{A_{NN}(t-s)} A_{NR} y_R(s) ds,$$

and substitute into $\dot{y}_R = A_{RR}y_R + A_{RN}y_N$. The convolution term has kernel $A_{RN}e^{A_{NN}(t-s)}A_{NR}$ and the boundary term is $F(t)$. Verified numerically: the GLE reconstructs the full resolved trajectory using only $(A_{RR}, A_{RN}, A_{NN}, A_{NR})$ and $y_N(0)$, to quadrature error. $\square$

Corollary 4.2 (The earlier “phase kernel” is the memory kernel). *If A_{NN} has complex spectrum, write its action as amplitude $\times$ phase: eigenvalues $\mu_k = -\gamma_k + i\omega_k$ give $e^{A_{NN}\tau} = \sum_k e^{-\gamma_k\tau} e^{i\omega_k\tau} \Pi_k$. Hence $K(\tau)$ has exactly the causal amplitude–phase form $K(\tau) \sim A(\tau)e^{i\Phi(\tau)}$, $\Phi(\tau) = \int_0^\tau \omega d\tau'$, of the CTMT “bona fide phase channel.” The earlier phase kernel is therefore the Mori–Zwanzig memory kernel of the null sector, with the phase set by the oscillatory part of the unresolved dynamics and the causal support built in.*

Remark 4.3 (Fluctuation–dissipation and colored residual). When $y_N(0)$ is drawn from the stationary null distribution, $F(t)$ is a zero-mean colored noise with $\mathbb{E}[F(t)F(s)^\top] = A_{RN}e^{A_{NN}t}\Sigma_N^0 e^{A_{NN}s}A_{RN}^\top$, tying the residual force to the kernel through the null covariance—the linear fluctuation–dissipation relation [6]. This is the honest home of the “residual F ” that appears heuristically in the earlier drafts.

Remark 4.4 (Physical instance: aftershocks as a memory response). The declustered seismic chart of the earlier CTMT seismic note is a concrete realisation. Independent mainshocks act as resolved forcing; the dependent aftershock sector is a driven relaxation response. Omori-type aftershock decay is precisely a (power-law) memory kernel: the unresolved/dependent sector is the integrated response of the resolved/independent forcing. This is exactly the $K(\tau)$ structure, and it is why the declustering projection Π_M of that note is a physical instance of the resolved/null split.

5 Directed information: what “buildup” must be measured by

To speak of buildup one needs a scalar that can increase. The correct, established choices are directed-information quantities, which for Gaussian linear systems are computable in closed form.

Proposition 5.1 (Coupling information in canonical correlations). *For jointly Gaussian $y = (y_R, y_N)$ the mutual information between sectors is*

$$I(y_R; y_N) = -\frac{1}{2} \log \det(I - \Sigma_R^{-1} C \Sigma_N^{-1} C^\top) = -\frac{1}{2} \sum_k \log(1 - \rho_k^2),$$

where ρ_k are the canonical correlations of the whitened coupling $\Sigma_R^{-1/2} C \Sigma_N^{-1/2}$ (the invariants of Paper III). It is nonnegative and vanishes iff $C = 0$.

Proof. Standard Gaussian mutual-information identity [9]; the canonical-correlation form is the singular value decomposition of the whitened coupling. $\square$

Proposition 5.2 (Directed coupling: Granger = transfer entropy). *For a stationary Gaussian linear system, the transfer entropy from the null to the resolved sector equals one half the Granger-causality $F_{N \rightarrow R}$ [7, 8], and $F_{N \rightarrow R} > 0$ iff the null sector improves one-step prediction of the resolved sector, which for Definition 2.1 holds iff $A_{RN} \neq 0$ (up to noise-coupling degeneracies).*

Propositions 5.1–5.2 give three natural *buildup functionals*: the sector mutual information $I(t) = I(y_R(t); y_N(t))$; the accumulated directed information / transfer entropy $N \rightarrow R$; and the static Fisher gain of Paper III promoted to a trajectory,

$$G(t) = J_R^\top (S_R(t)^{-1} - \Sigma_R(t)^{-1}) J_R, \quad S_R(t) = \Sigma_R(t) - C(t) \Sigma_N(t)^{-1} C(t)^\top.$$

Each is ≥ 0 and vanishes iff $C(t) = 0$. Whether any is *monotone* along the Lyapunov flow is the content of the ladder below; it is not automatic.

6 Why Fisher is the forced metric (Čencov anchor)

The choice of Fisher/whitening metric in Papers I–II is not a convention. Čencov’s theorem states that, up to scale, the Fisher metric is the unique Riemannian metric on a statistical model invariant under Markov morphisms (sufficient statistics) [10, 11, 12]. This is the external justification for using $g = \Sigma^{-1}$ as the reference metric and for treating $J^\top \Sigma^{-1} J$ as the canonical response metric: any admissible coarse-graining of observations must respect it. The earlier “forced admissibility class” (Fisher-admissible $\cap$ inversion-stable $\cap$ dimensionally-closed) is the correct conservative use of this fact, once exact kernel equalities are replaced by regularized stability bounds, as in the Tikhonov formulation of that draft.

7 Atlas to bundle: connection, curvature, monodromy

When the operating point θ varies over a base manifold Θ , the resolved sector $R(\theta)$ is a moving subspace: a subbundle of the trivial bundle $O \times \Theta$, i.e. a map $\theta \mapsto P_R(\theta)$ into the Grassmannian. This upgrades the static Čech atlas of Paper III to a bundle with a connection.

Definition 7.1 (Projection connection and curvature). On the resolved subbundle, the projection (Grassmann) connection is $D = P_R \circ d$; its curvature is the $\text{Aut}(\mathcal{G})$ -valued two-form $\mathcal{F} = P_R(dP_R) \wedge (dP_R) P_R$. The holonomy of D around a loop $\gamma \subset \Theta$ is $\text{Hol}_\gamma \in \text{Aut}(\mathcal{G})$.

Proposition 7.2 (Monodromy is holonomy; Ambrose–Singer). *The transport monodromy computed in the earlier CTMT seismic note—rolling loop products $H_\gamma = \prod T_{\alpha\beta}$ of estimated inter-chart transitions—is a discrete estimator of Hol_γ . By the Ambrose–Singer theorem, the holonomy algebra is generated by the curvature $\mathcal{F}$ transported over Θ [13, 14]. Nontrivial curvature is therefore the infinitesimal source of the nontrivial loop defects observed empirically.*

The curvature $\mathcal{F}$ measures the failure of the resolved distribution to be integrable—how much R twists into N under transport. This is *geometric* seepage, and it is a priori distinct from the *dynamical* seepage A_{RN} of Section 3. Their relationship is the last rung of the ladder.

8 The ladder: what remains to be proved

We state the open problems as an ordered ladder, each rung with its hypotheses and the established tools that should settle it. This is the paper’s roadmap; none of these is claimed as proved.

Conjecture 8.1 (Rung 1: monotone buildup near a decoupled state). *Let the flow start decoupled ($C(0) = 0$, $Q_{RN} = 0$) with $A_{RN}, A_{NR} \neq 0$. Then on a nonempty interval $[0, t_*)$ the sector mutual information $I(t)$ (Proposition 5.1) is strictly increasing. Tools: $\dot{I}(0) = 0$, and $\ddot{I}(0) > 0$ should follow from Theorem 3.2 since $C(t) = t(A_{RN}\Sigma_N + \Sigma_R A_{NR}^\top) + O(t^2)$ makes $I(t) = \frac{1}{2}\|\Sigma_R^{-1/2}C\Sigma_N^{-1/2}\|_F^2 + o(t^2) \sim t^2$. This local statement is the first provable buildup result; prove it, then seek the maximal t_* .*

Conjecture 8.2 (Rung 2: global buildup conditions). *Characterise the pairs (A, Q) for which a chosen buildup functional ($I(t)$, accumulated transfer entropy, or $G(t)$) is monotone on all of $[0, \infty)$ up to the stationary value, versus those for which it overshoots. Tools: the explicit Lyapunov solution $\Sigma(t) = \int_0^t e^{As} Q e^{A^\top s} ds$ under $\Sigma(0) = 0$; monotone-operator and Loewner-order techniques.*

Conjecture 8.3 (Rung 3: kernel-controlled buildup). *Bound the stationary coupling $\|C_\infty\|$ and the buildup rate in terms of the memory kernel norm $\int_0^\infty \|K(\tau)\| d\tau$ and the null relaxation $\|e^{A_{NN}\tau}\|$. Tools: Theorem 4.1, fluctuation–dissipation, and standard convolution (Young) inequalities.*

Conjecture 8.4 (Rung 4: curvature = dynamical seepage). *Under a compatibility condition relating the base flow on Θ to the drift A (the parameter motion is generated by the same dynamics), the projection curvature $\mathcal{F}$ of Section 7 is proportional to the cross-drift A_{RN} (equivalently to the memory kernel K). Tools: adiabatic/Berry transport of the resolved eigenspace; Ambrose–Singer. This is the rung that would make “the atlas curvature is the seepage” a theorem rather than an analogy, and would complete the passage from Paper III’s atlas to a genuine gauge theory of resolution geometry.*

Conjecture 8.5 (Rung 5: canonical connection). *Show the connection is forced (not chosen) by the fluctuation–dissipation structure, so that the curvature is an invariant of the dynamical resolution geometry rather than a gauge artefact. Tools: Čencov-type uniqueness (Section 6) transported to the bundle setting.*

Conjecture 8.6 (Rung 6: singular/rank-changing globalization). *Extend all of the above to charts where $\text{rank } J$ or $\text{range } \Sigma$ changes (Paper II singular results), classifying the stratified obstructions.*

9 Falsification protocol

The dynamical layer is testable on data (synthetic or real) by the defect-and- envelope methodology of the earlier coil battery, now applied to the generator.

Program step 9.1 (Estimate the generator). From a trajectory of the full observable y_t , identify A and Q (linear system identification / regression of $\dot{y}$ on y ; residual covariance for Q). The full state is observed; the null components carry no first-order parameter signal but are measured, so A including A_{RN}, A_{NR} is identifiable.

Program step 9.2 (Test for seepage). Split by the Jacobian into $R \oplus N$. Report $\|A_{RN}\|, \|A_{NR}\|$ and the memory kernel $K(\tau) = A_{RN} e^{A_{NN}\tau} A_{NR}$; test the Granger/transfer-entropy $F_{N \rightarrow R}$ (Proposition 5.2) against a null of no directed coupling.

Program step 9.3 (Test for buildup). Track a buildup functional ($I(t)$, accumulated transfer entropy, or $G(t)$) from a near-decoupled initial condition; check monotonicity against a preregistered tolerance (Rung 1). Report failures honestly—the coil battery’s lesson was that added stresses need not sharpen the class, and a negative result is informative.

Program step 9.4 (Test the connection). On a moving operating point, compute inter-chart transitions and loop monodromy (as in the seismic note); compare to the projection holonomy and estimate the curvature. Nontrivial, gauge-stable curvature is the signature sought by Rung 4.

10 Non-claims and conclusion

Non-claim 10.1 (Mechanism, not phenomenon). This paper proves that seepage has a generation term (Theorem 3.2) and a memory kernel (Theorem 4.1), and that the earlier phase kernel is that memory kernel (Corollary 4.2). It does not prove that buildup is monotone (Rungs 1–3), that curvature equals seepage (Rungs 4–5), or that any physical system realises the required drift structure. Those are the stated open problems.

The static theory could not accommodate accumulation because it contained no generator. Adding the minimal established generator—a linear stochastic evolution with its Lyapunov equation—converts the CTMT thesis into mathematics that is either provable or refutable. Seepage becomes the generation term $A_{RN}\Sigma_N + \Sigma_R A_{NR}^\top$ and the Mori–Zwanzig kernel $A_{RN}e^{A_{NN}^\top}A_{NR}$; the “phase kernel” invention becomes a known object; directed coupling is measured by Gaussian transfer entropy; and the atlas becomes a bundle whose curvature is the candidate for geometric seepage. What remains—the monotonicity of buildup and the curvature identification—is now a finite, ordered ladder of well-posed problems, each with its standard tool, rather than an interpretation to be defended. That is the honest boundary of the present stage: the mechanism is proved; the accumulation is the next theorem.

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